Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

DAIRY FACILITY INFORMATION

A. NAME OF DAIRY OR BUSINESS OPERATING THE DAIRY: Temple Creek Dairy, Inc.

Physical address of dairy:

26250 Carter Road	Escalon	San Joaquin	95320
Number and Street	City	County	Zip Code

Street and nearest cross street (if no address):

Date facility was originally placed in operation: 02/01/1994

Regional Water Quality Control Board Basin Plan designation: San Joaquin River Basin

County Assessor Parcel Number(s) for dairy facility:

X207-X040-X001-XXXX

B. OPERATORS

Veneman, Robert Eric

Operator name: Veneman, Robert Eric

Telephone no.: (123) 000-0000 (Landline) (Cellular)

26250 Carter RD	Escalon	CA	95320
Mailing Address Number and Street	City	State	Zip Code

This operator is responsible for paying permit fees.

C. OWNERS

Veneman, Robert Eric

Legal owner name: Veneman, Robert Eric

Telephone no.: (123) 000-0000 (Landline) (Cellular)

26250 Carter RD	Escalon	CA	95320
Mailing Address Number and Street	City	State	Zip Code

This owner is responsible for paying permit fees.

AVAILABLE NUTRIENTS

A. HERD INFORMATION

	Milk Cows	Dry Cows	Bred Heifers (15-24 mo.)	Heifers (7-14 mo. to breeding)	Calves (4-6 mo.)	Calves (0-3 mo.)
Number open confinement	0	0	800	885	420	450
Number under roof	2,755	250	0	0	0	0
Maximum number	2,790	350	950	975	545	600
Average number	2,755	250	800	885	420	450
Avg live weight (lbs)	1,300	1,350	1,100	75		

Predominant milk cow breed: Holstein

Average milk production: 85 *pounds per cow per day*

B. MANURE GENERATED

Total manure excreted by the herd: 97,306.27 *tons per reporting period*

Total nitrogen from manure: 1,237,372.23 *lbs per reporting period*

After ammonia losses (30% loss applied): 866,160.56 *lbs per reporting period*

Total phosphorus from manure: 204,309.92 *lbs per reporting period*

Total potassium from manure: 554,758.44 *lbs per reporting period*

Total salt from manure: 1,354,679.25 *lbs per reporting period*

C. PROCESS WASTEWATER GENERATED

Process wastewater generated: 28,841,480 *gallons*

Total nitrogen generated: 77,027.96 *lbs*

Total phosphorus generated: 6,986.22 *lbs*

Total potassium generated: 101,645.68 *lbs*

Total salt generated: 543,701.85 *lbs*

	28,841,480	*gallons applied*
+	0	*gallons exported*
-	0	*gallons imported*
=	28,841,480	*gallons generated*

D. FRESH WATER SOURCES

Source Description	Type
Ag Well #1 Office	Ground water
Ag Well #2 East	Ground water
Ag Well #3 South	Ground water
Domestic Well	Ground water

E. SUBSURFACE (TILE) DRAINAGE SOURCES

No subsurface (tile) drainage sources entered.

F. NUTRIENT IMPORTS

No dry manure nutrient imports entered.

No process wastewater nutrient imports entered.

No commercial or other nutrient imports entered.

G. NUTRIENT EXPORTS

Date	Material type	Quantity	Reporting basis	Moisture (%)	Density (lbs/cu ft)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
06/26/2023	Corral solids	2,500.00 *ton*	Dry-weight	0.1		0.00	0.00	0.00		0.00

No liquid nutrient exports entered.

Material type	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Total exports for all materials	0.00	0.00	0.00	0.00

APPLICATION AREA

A. LIST OF LAND APPLICATION AREAS

Field name	Controlled acres	Cropable acres	Total harvests	Type of waste applied	Parcel number
Almonds	346	346	0	none	X002-X002-X002-XXXX
Home	4	4	0	none	X004-X075-X046-XXXX
TC1	67	67	2	process wastewater	X207-X040-X003-XXXX
TC10	59	59	2	both	X207-X040-X003-XXXX
TC2	28	28	1	process wastewater	X207-X040-X003-XXXX
TC3	26	26	1	process wastewater	X207-X040-X002-XXXX
TC4	27	27	1	process wastewater	X207-X040-X002-XXXX
TC5	30	30	2	process wastewater	X207-X040-X002-XXXX
TC6	15	15	1	process wastewater	X207-X040-X002-XXXX
TC7	88	88	2	process wastewater	X207-X040-X003-XXXX
TC8	89	89	2	process wastewater	X207-X040-X003-XXXX
TC9	15	15	1	process wastewater	X207-X040-X003-XXXX
TC9A	50	50	2	both	X207-X040-X003-XXXX
Totals for areas that were used for application	494	494	17		
Totals for areas that were not used for application	350	350	0		
Land application area totals	844	844	17		

B. CROPS AND HARVESTS

TC1

Field name: TC1

10/24/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 67 Plant date: 10/24/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/10/2023	938.00 *ton*	Dry-weight		62.2	26,700.00	1,940.00	21,300.00		14.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	14.00	282.59	20.53	225.44	1,481.76

TC1

06/15/2023: Corn, silage

Crop: Corn, silage Acres planted: 67 Plant date: 06/15/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/10/2023	1,742.00 *ton*	Dry-weight		0.1	0.00	0.00	0.00		0.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	26.00	0.00	0.00	0.00	0.00

TC10

Field name: TC10

10/22/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 59 Plant date: 10/22/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/14/2023	826.00 *ton*	Dry-weight		77.5	34,400.00	2,650.00	32,200.00		15.10

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	14.00	216.72	16.70	202.86	951.30

06/15/2023: Corn, silage

Crop: Corn, silage Acres planted: 59 Plant date: 06/15/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/16/2023	1,534.00 *ton*	Dry-weight		0.1	0.00	0.00	0.00		0.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	26.00	0.00	0.00	0.00	0.00

TC2

Field name: TC2

06/16/2023: Corn, silage

Crop: Corn, silage Acres planted: 28 Plant date: 06/16/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/10/2023	728.00 *ton*	Dry-weight		0.1	0.00	0.00	0.00		0.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	26.00	0.00	0.00	0.00	0.00

TC3

Field name: TC3

06/16/2023: Corn, silage

Crop: Corn, silage Acres planted: 26 Plant date: 06/16/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/10/2023	676.00 *ton*	Dry-weight		0.1	0.00	0.00	0.00		0.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	26.00	0.00	0.00	0.00	0.00

TC4

Field name: TC4

TC4

06/17/2023: Corn, silage

Crop: Corn, silage Acres planted: 27 Plant date: 06/17/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/11/2023	702.00 *ton*	Dry-weight		0.1	0.00	0.00	0.00		0.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	26.00	0.00	0.00	0.00	0.00

TC5

Field name: TC5

10/26/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 30 Plant date: 10/26/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/12/2023	420.00 *ton*	Dry-weight		58.3	27,500.00	1,590.00	19,400.00		12.60

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	14.00	321.09	18.56	226.51	1,471.18

06/17/2023: Corn, silage

Crop: Corn, silage Acres planted: 30 Plant date: 06/17/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/11/2023	780.00 *ton*	Dry-weight		0.1	0.00	0.00	0.00		0.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	26.00	0.00	0.00	0.00	0.00

TC6

Field name: TC6

06/18/2023: Corn, silage

Crop: Corn, silage Acres planted: 15 Plant date: 06/18/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/12/2023	390.00 *ton*	Dry-weight		0.1	0.00	0.00	0.00		0.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	26.00	0.00	0.00	0.00	0.00

TC7

Field name: TC7

10/24/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 88 Plant date: 10/24/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/12/2023	1,232.00 *ton*	Dry-weight		59.5	22,900.00	2,320.00	34,200.00		12.20

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	14.00	259.69	26.31	387.83	1,383.48

06/18/2023: Corn, silage

Crop: Corn, silage Acres planted: 88 Plant date: 06/18/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/12/2023	2,288.00 *ton*	Dry-weight		0.1	0.00	0.00	0.00		0.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	26.00	0.00	0.00	0.00	0.00

TC8

Field name: TC8

10/23/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 89 Plant date: 10/23/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/13/2023	1,246.00 *ton*	Dry-weight		68.4	24,900.00	2,230.00	20,400.00		12.20

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	14.00	220.32	19.73	180.50	1,079.46

06/19/2023: Corn, silage

Crop: Corn, silage Acres planted: 89 Plant date: 06/19/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/15/2023	2,314.00 *ton*	Dry-weight		0.1	0.00	0.00	0.00		0.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	26.00	0.00	0.00	0.00	0.00

TC9

Field name: TC9

06/20/2023: Corn, silage

Crop: Corn, silage Acres planted: 15 Plant date: 06/20/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/14/2023	390.00 *ton*	Dry-weight		0.1	0.00	0.00	0.00		0.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	26.00	0.00	0.00	0.00	0.00

TC9A

Field name: TC9A

10/22/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 50 Plant date: 10/22/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/14/2023	700.00 *ton*	Dry-weight		81.5	36,600.00	1,730.00	19,800.00		15.70

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	14.00	189.59	8.96	102.56	813.26

06/21/2023: Corn, silage

Crop: Corn, silage Acres planted: 50 Plant date: 06/21/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/14/2023	1,300.00 *ton*	Dry-weight		0.1	0.00	0.00	0.00		0.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	26.00	0.00	0.00	0.00	0.00

NUTRIENT BUDGET

A. LAND APPLICATIONS

TC1 - 10/24/2022: Oats, silage-soft dough

Field name: TC1

Crop: Oats, silage-soft dough

Plant date: 10/24/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
01/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	101.59	7.30	130.43	753.29	1,728,000.00 *gal*
Application event totals		101.59	7.30	130.43	753.29	

TC1 - 06/15/2023: Corn, silage

Field name: TC1

Crop: Corn, silage

Plant date: 06/15/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well #1 Office	Ground water	0.26	0.00	0.00	26.90	1,200,000.00 *gal*
Ag Well #2 East	Ground water	0.27	0.00	0.00	251.10	1,260,000.00 *gal*
Ag Well #3 South	Ground water	0.76	0.00	0.00	30.64	1,230,000.00 *gal*
Application event totals		1.29	0.00	0.00	308.64	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well #1 Office	Ground water	0.26	0.00	0.00	26.90	1,200,000.00 *gal*
Ag Well #2 East	Ground water	0.27	0.00	0.00	251.10	1,260,000.00 *gal*
Ag Well #3 South	Ground water	0.76	0.00	0.00	30.64	1,230,000.00 *gal*
Application event totals		1.29	0.00	0.00	308.64	

TC1 - 06/15/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	5.97	1.45	12.14	66.29	691,200.00 *gal*
Ag Well #1 Office	Ground water	0.12	0.00	0.00	12.91	576,000.00 *gal*
Ag Well #2 East	Ground water	0.13	0.00	0.00	120.53	604,800.00 *gal*
Ag Well #3 South	Ground water	0.37	0.00	0.00	14.71	590,400.00 *gal*
Application event totals		6.60	1.45	12.14	214.44	

08/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	5.97	1.45	12.14	66.29	691,200.00 *gal*
Ag Well #1 Office	Ground water	0.12	0.00	0.00	12.91	576,000.00 *gal*
Ag Well #2 East	Ground water	0.13	0.00	0.00	120.53	604,800.00 *gal*
Ag Well #3 South	Ground water	0.37	0.00	0.00	14.71	590,400.00 *gal*
Application event totals		6.60	1.45	12.14	214.44	

08/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	43.33	3.41	51.37	236.75	576,000.00 *gal*
Ag Well #1 Office	Ground water	0.10	0.00	0.00	10.76	480,000.00 *gal*
Ag Well #2 East	Ground water	0.11	0.00	0.00	100.44	504,000.00 *gal*
Ag Well #3 South	Ground water	0.31	0.00	0.00	12.26	492,000.00 *gal*
Application event totals		43.85	3.41	51.37	360.21	

09/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well #1 Office	Ground water	0.17	0.00	0.00	17.92	799,200.00 *gal*
Ag Well #2 East	Ground water	0.18	0.00	0.00	167.23	839,160.00 *gal*
Ag Well #3 South	Ground water	0.51	0.00	0.00	20.41	819,180.00 *gal*
Application event totals		0.86	0.00	0.00	205.55	

TC10 - 10/22/2022: Oats, silage-soft dough

TC10 - 10/22/2022: Oats, silage-soft dough

Field name: TC10

Crop: Oats, silage-soft dough

Plant date: 10/22/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/08/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Manure	Corral solids	240.04	81.64	200.56	6,522.10	708.00 *ton*
Application event totals		240.04	81.64	200.56	6,522.10	

TC10 - 06/15/2023: Corn, silage

Field name: TC10

Crop: Corn, silage

Plant date: 06/15/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well #1 Office	Ground water	0.35	0.00	0.00	36.66	1,440,000.00 *gal*
Ag Well #2 East	Ground water	0.37	0.00	0.00	342.17	1,512,000.00 *gal*
Ag Well #3 South	Ground water	1.04	0.00	0.00	41.75	1,476,000.00 *gal*
Application event totals		1.76	0.00	0.00	420.59	

07/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well #2 East	Ground water	0.37	0.00	0.00	342.17	1,512,000.00 *gal*
Ag Well #3 South	Ground water	1.04	0.00	0.00	41.75	1,476,000.00 *gal*
Ag Well #1 Office	Ground water	0.35	0.00	0.00	36.66	1,440,000.00 *gal*
Application event totals		1.76	0.00	0.00	420.59	

TC10 - 06/15/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	9.61	2.34	19.53	106.64	979,200.00 *gal*
Ag Well #1 Office	Ground water	0.20	0.00	0.00	20.77	816,000.00 *gal*
Ag Well #2 East	Ground water	0.21	0.00	0.00	193.90	856,800.00 *gal*
Ag Well #3 South	Ground water	0.59	0.00	0.00	23.66	836,400.00 *gal*
Application event totals		10.61	2.34	19.53	344.98	

08/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	9.61	2.34	19.53	106.64	979,200.00 *gal*
Ag Well #1 Office	Ground water	0.20	0.00	0.00	20.77	816,000.00 *gal*
Ag Well #2 East	Ground water	0.21	0.00	0.00	193.90	856,800.00 *gal*
Ag Well #3 South	Ground water	0.59	0.00	0.00	23.66	836,400.00 *gal*
Application event totals		10.61	2.34	19.53	344.98	

08/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	83.65	6.58	99.17	457.05	979,200.00 *gal*
Ag Well #1 Office	Ground water	0.20	0.00	0.00	20.77	816,000.00 *gal*
Ag Well #2 East	Ground water	0.21	0.00	0.00	193.90	856,800.00 *gal*
Ag Well #3 South	Ground water	0.59	0.00	0.00	23.66	836,400.00 *gal*
Application event totals		84.65	6.58	99.17	695.38	

09/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well #1 Office	Ground water	0.31	0.00	0.00	32.57	1,279,200.00 *gal*
Ag Well #2 East	Ground water	0.33	0.00	0.00	303.96	1,343,160.00 *gal*
Ag Well #3 South	Ground water	0.93	0.00	0.00	37.09	1,311,180.00 *gal*
Application event totals		1.57	0.00	0.00	373.62	

TC2 - 06/16/2023: Corn, silage

TC2 - 06/16/2023: Corn, silage

Field name: TC2

Crop: Corn, silage

Plant date: 06/16/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well #1 Office	Ground water	0.31	0.00	0.00	32.19	600,000.00 *gal*
Ag Well #2 East	Ground water	0.32	0.00	0.00	300.42	630,000.00 *gal*
Ag Well #3 South	Ground water	0.91	0.00	0.00	36.66	615,000.00 *gal*
Application event totals		1.55	0.00	0.00	369.27	

07/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well #1 Office	Ground water	0.31	0.00	0.00	32.19	600,000.00 *gal*
Ag Well #2 East	Ground water	0.32	0.00	0.00	300.42	630,000.00 *gal*
Ag Well #3 South	Ground water	0.91	0.00	0.00	36.66	615,000.00 *gal*
Application event totals		1.55	0.00	0.00	369.27	

07/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	8.34	2.03	16.94	92.53	403,200.00 *gal*
Ag Well #1 Office	Ground water	0.17	0.00	0.00	18.03	336,000.00 *gal*
Ag Well #2 East	Ground water	0.18	0.00	0.00	168.24	352,800.00 *gal*
Ag Well #3 South	Ground water	0.51	0.00	0.00	20.53	344,400.00 *gal*
Application event totals		9.21	2.03	16.94	299.32	

08/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	8.34	2.03	16.94	92.53	403,200.00 *gal*
Ag Well #1 Office	Ground water	0.17	0.00	0.00	18.03	336,000.00 *gal*
Ag Well #2 East	Ground water	0.18	0.00	0.00	168.24	352,800.00 *gal*
Ag Well #3 South	Ground water	0.51	0.00	0.00	20.53	344,400.00 *gal*
Application event totals		9.21	2.03	16.94	299.32	

TC2 - 06/16/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	72.58	5.71	86.04	396.55	403,200.00 *gal*
Ag Well #1 Office	Ground water	0.17	0.00	0.00	18.03	336,000.00 *gal*
Ag Well #2 East	Ground water	0.18	0.00	0.00	168.24	352,800.00 *gal*
Application event totals		72.94	5.71	86.04	582.81	

09/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well #1 Office	Ground water	0.25	0.00	0.00	25.75	480,000.00 *gal*
Ag Well #2 East	Ground water	0.26	0.00	0.00	240.34	504,000.00 *gal*
Ag Well #3 South	Ground water	0.73	0.00	0.00	29.33	492,000.00 *gal*
Application event totals		1.24	0.00	0.00	295.41	

TC3 - 06/16/2023: Corn, silage

Field name: TC3

Crop: Corn, silage

Plant date: 06/16/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well #1 Office	Ground water	0.30	0.00	0.00	31.89	552,000.00 *gal*
Ag Well #2 East	Ground water	0.32	0.00	0.00	297.65	579,600.00 *gal*
Ag Well #3 South	Ground water	0.91	0.00	0.00	36.32	565,800.00 *gal*
Application event totals		1.53	0.00	0.00	365.86	

TC3 - 06/16/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well #1 Office	Ground water	0.30	0.00	0.00	31.89	552,000.00 *gal*
Ag Well #2 East	Ground water	0.32	0.00	0.00	297.65	579,600.00 *gal*
Ag Well #3 South	Ground water	0.91	0.00	0.00	36.32	565,800.00 *gal*
Application event totals		1.53	0.00	0.00	365.86	

07/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	5.21	1.27	10.59	57.83	234,000.00 *gal*
Ag Well #1 Office	Ground water	0.17	0.00	0.00	18.03	312,000.00 *gal*
Ag Well #2 East	Ground water	0.18	0.00	0.00	168.24	327,600.00 *gal*
Ag Well #3 South	Ground water	0.51	0.00	0.00	20.53	319,800.00 *gal*
Application event totals		6.08	1.27	10.59	264.62	

08/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	5.21	1.27	10.59	57.83	234,000.00 *gal*
Ag Well #2 East	Ground water	0.18	0.00	0.00	168.24	327,600.00 *gal*
Ag Well #3 South	Ground water	0.51	0.00	0.00	20.53	319,800.00 *gal*
Ag Well #1 Office	Ground water	0.17	0.00	0.00	18.03	312,000.00 *gal*
Application event totals		6.08	1.27	10.59	264.62	

08/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	45.36	3.57	53.78	247.85	234,000.00 *gal*
Ag Well #1 Office	Ground water	0.17	0.00	0.00	18.03	312,000.00 *gal*
Ag Well #2 East	Ground water	0.18	0.00	0.00	168.24	327,600.00 *gal*
Ag Well #3 South	Ground water	0.51	0.00	0.00	20.53	319,800.00 *gal*
Application event totals		46.23	3.57	53.78	454.64	

TC3 - 06/16/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well #1 Office	Ground water	0.24	0.00	0.00	25.37	439,200.00 *gal*
Ag Well #2 East	Ground water	0.26	0.00	0.00	236.82	461,160.00 *gal*
Ag Well #3 South	Ground water	0.72	0.00	0.00	28.90	450,180.00 *gal*
Application event totals		1.22	0.00	0.00	291.10	

TC4 - 06/17/2023: Corn, silage

Field name: TC4

Crop: Corn, silage

Plant date: 06/17/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well #1 Office	Ground water	0.31	0.00	0.00	32.25	579,600.00 *gal*
Ag Well #2 East	Ground water	0.33	0.00	0.00	300.95	608,580.00 *gal*
Ag Well #3 South	Ground water	0.92	0.00	0.00	36.72	594,090.00 *gal*
Application event totals		1.55	0.00	0.00	369.92	

07/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well #1 Office	Ground water	0.31	0.00	0.00	32.25	579,600.00 *gal*
Ag Well #2 East	Ground water	0.33	0.00	0.00	300.95	608,580.00 *gal*
Ag Well #3 South	Ground water	0.92	0.00	0.00	36.72	594,090.00 *gal*
Application event totals		1.55	0.00	0.00	369.92	

TC4 - 06/17/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	5.21	1.27	10.59	57.83	243,000.00 *gal*
Ag Well #1 Office	Ground water	0.17	0.00	0.00	18.03	324,000.00 *gal*
Ag Well #2 East	Ground water	0.18	0.00	0.00	168.24	340,200.00 *gal*
Ag Well #3 South	Ground water	0.51	0.00	0.00	20.53	332,100.00 *gal*
Application event totals		6.08	1.27	10.59	264.62	

08/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	5.21	1.27	10.59	57.83	243,000.00 *gal*
Ag Well #1 Office	Ground water	0.17	0.00	0.00	18.03	324,000.00 *gal*
Ag Well #2 East	Ground water	0.18	0.00	0.00	168.24	340,200.00 *gal*
Ag Well #3 South	Ground water	0.51	0.00	0.00	20.53	332,100.00 *gal*
Application event totals		6.08	1.27	10.59	264.62	

08/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	45.36	3.57	53.78	247.85	243,000.00 *gal*
Ag Well #1 Office	Ground water	0.17	0.00	0.00	18.03	324,000.00 *gal*
Ag Well #2 East	Ground water	0.18	0.00	0.00	168.24	340,200.00 *gal*
Ag Well #3 South	Ground water	0.51	0.00	0.00	20.53	332,100.00 *gal*
Application event totals		46.23	3.57	53.78	454.64	

09/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well #1 Office	Ground water	0.24	0.00	0.00	25.57	459,600.00 *gal*
Ag Well #2 East	Ground water	0.26	0.00	0.00	238.64	482,580.00 *gal*
Ag Well #3 South	Ground water	0.73	0.00	0.00	29.12	471,090.00 *gal*
Application event totals		1.23	0.00	0.00	293.33	

TC5 - 10/26/2022: Oats, silage-soft dough

TC5 - 10/26/2022: Oats, silage-soft dough

Field name: TC5

Crop: Oats, silage-soft dough

Plant date: 10/26/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/01/2022	Injection	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	331.82	23.83	426.02	2,460.52	2,527,280.00 *gal*
Application event totals		331.82	23.83	426.02	2,460.52	

TC5 - 06/17/2023: Corn, silage

Field name: TC5

Crop: Corn, silage

Plant date: 06/17/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well #1 Office	Ground water	0.32	0.00	0.00	34.01	679,200.00 *gal*
Ag Well #2 East	Ground water	0.34	0.00	0.00	317.40	713,160.00 *gal*
Ag Well #3 South	Ground water	0.97	0.00	0.00	38.73	696,180.00 *gal*
Application event totals		1.63	0.00	0.00	390.14	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well #1 Office	Ground water	0.32	0.00	0.00	34.01	679,200.00 *gal*
Ag Well #2 East	Ground water	0.34	0.00	0.00	317.40	713,160.00 *gal*
Ag Well #3 South	Ground water	0.97	0.00	0.00	38.73	696,180.00 *gal*
Application event totals		1.63	0.00	0.00	390.14	

TC5 - 06/17/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	8.90	2.17	18.07	98.70	460,800.00 *gal*
Ag Well #1 Office	Ground water	0.18	0.00	0.00	19.23	384,000.00 *gal*
Ag Well #2 East	Ground water	0.19	0.00	0.00	179.45	403,200.00 *gal*
Ag Well #3 South	Ground water	0.55	0.00	0.00	21.90	393,600.00 *gal*
Application event totals		9.82	2.17	18.07	319.27	

08/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	8.90	2.17	18.07	98.70	460,800.00 *gal*
Ag Well #1 Office	Ground water	0.18	0.00	0.00	19.23	384,000.00 *gal*
Ag Well #2 East	Ground water	0.19	0.00	0.00	179.45	403,200.00 *gal*
Ag Well #3 South	Ground water	0.55	0.00	0.00	21.90	393,600.00 *gal*
Application event totals		9.82	2.17	18.07	319.27	

08/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	8.90	2.17	18.07	98.70	460,800.00 *gal*
Ag Well #1 Office	Ground water	0.18	0.00	0.00	19.23	384,000.00 *gal*
Ag Well #2 East	Ground water	0.19	0.00	0.00	179.45	403,200.00 *gal*
Ag Well #3 South	Ground water	0.55	0.00	0.00	21.90	393,600.00 *gal*
Application event totals		9.82	2.17	18.07	319.27	

09/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well #1 Office	Ground water	0.27	0.00	0.00	28.00	559,200.00 *gal*
Ag Well #2 East	Ground water	0.28	0.00	0.00	261.33	587,160.00 *gal*
Ag Well #3 South	Ground water	0.80	0.00	0.00	31.89	573,180.00 *gal*
Application event totals		1.35	0.00	0.00	321.21	

TC6 - 06/18/2023: Corn, silage

TC6 - 06/18/2023: Corn, silage

Field name: TC6

Crop: Corn, silage

Plant date: 06/18/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well #1 Office	Ground water	0.52	0.00	0.00	54.08	540,000.00 *gal*
Ag Well #2 East	Ground water	0.55	0.00	0.00	504.71	567,000.00 *gal*
Application event totals		1.06	0.00	0.00	558.78	

07/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well #1 Office	Ground water	0.52	0.00	0.00	54.08	540,000.00 *gal*
Ag Well #2 East	Ground water	0.55	0.00	0.00	504.71	567,000.00 *gal*
Application event totals		1.06	0.00	0.00	558.78	

07/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	5.56	1.35	11.30	61.69	144,000.00 *gal*
Ag Well #1 Office	Ground water	0.28	0.00	0.00	28.84	288,000.00 *gal*
Ag Well #2 East	Ground water	0.29	0.00	0.00	269.18	302,400.00 *gal*
Application event totals		6.13	1.35	11.30	359.70	

08/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	5.56	1.35	11.30	61.69	144,000.00 *gal*
Ag Well #1 Office	Ground water	0.28	0.00	0.00	28.84	288,000.00 *gal*
Ag Well #2 East	Ground water	0.29	0.00	0.00	269.18	302,400.00 *gal*
Application event totals		6.13	1.35	11.30	359.70	

TC6 - 06/18/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	48.39	3.81	57.36	264.37	144,000.00 *gal*
Ag Well #1 Office	Ground water	0.28	0.00	0.00	28.84	288,000.00 *gal*
Ag Well #2 East	Ground water	0.29	0.00	0.00	269.18	302,400.00 *gal*
Application event totals		48.95	3.81	57.36	562.39	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well #1 Office	Ground water	0.40	0.00	0.00	42.06	420,000.00 *gal*
Ag Well #2 East	Ground water	0.42	0.00	0.00	392.55	441,000.00 *gal*
Application event totals		0.83	0.00	0.00	434.61	

TC7 - 10/24/2022: Oats, silage-soft dough

Field name: TC7

Crop: Oats, silage-soft dough Plant date: 10/24/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
01/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	128.91	9.26	165.50	955.88	2,880,000.00 *gal*
Application event totals		128.91	9.26	165.50	955.88	

TC7 - 06/18/2023: Corn, silage

Field name: TC7

Crop: Corn, silage Plant date: 06/18/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

TC7 - 06/18/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well #1 Office	Ground water	0.23	0.00	0.00	24.58	1,440,000.00 *gal*
Ag Well #2 East	Ground water	0.25	0.00	0.00	229.41	1,512,000.00 *gal*
Ag Well #3 South	Ground water	0.70	0.00	0.00	27.99	1,476,000.00 *gal*
Application event totals		1.18	0.00	0.00	281.99	

07/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well #1 Office	Ground water	0.23	0.00	0.00	24.58	1,440,000.00 *gal*
Ag Well #2 East	Ground water	0.25	0.00	0.00	229.41	1,512,000.00 *gal*
Ag Well #3 South	Ground water	0.70	0.00	0.00	27.99	1,476,000.00 *gal*
Application event totals		1.18	0.00	0.00	281.99	

07/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	6.07	1.48	12.32	67.29	921,600.00 *gal*
Ag Well #1 Office	Ground water	0.13	0.00	0.00	13.11	768,000.00 *gal*
Ag Well #2 East	Ground water	0.13	0.00	0.00	122.35	806,400.00 *gal*
Ag Well #3 South	Ground water	0.37	0.00	0.00	14.93	787,200.00 *gal*
Application event totals		6.70	1.48	12.32	217.69	

08/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	6.07	1.48	12.32	67.29	921,600.00 *gal*
Ag Well #1 Office	Ground water	0.13	0.00	0.00	13.11	768,000.00 *gal*
Ag Well #2 East	Ground water	0.13	0.00	0.00	122.35	806,400.00 *gal*
Ag Well #3 South	Ground water	0.37	0.00	0.00	14.93	787,200.00 *gal*
Application event totals		6.70	1.48	12.32	217.69	

TC7 - 06/18/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	52.79	4.15	62.57	288.40	921,600.00 *gal*
Ag Well #1 Office	Ground water	0.13	0.00	0.00	13.11	768,000.00 *gal*
Ag Well #2 East	Ground water	0.13	0.00	0.00	122.35	806,400.00 *gal*
Ag Well #3 South	Ground water	0.37	0.00	0.00	14.93	787,200.00 *gal*
Application event totals		53.42	4.15	62.57	438.80	

09/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well #1 Office	Ground water	0.20	0.00	0.00	20.48	1,200,000.00 *gal*
Ag Well #2 East	Ground water	0.21	0.00	0.00	191.18	1,260,000.00 *gal*
Ag Well #3 South	Ground water	0.58	0.00	0.00	23.33	1,230,000.00 *gal*
Application event totals		0.98	0.00	0.00	234.99	

TC8 - 10/23/2022: Oats, silage-soft dough

Field name: TC8

Crop: Oats, silage-soft dough Plant date: 10/23/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
01/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	127.46	9.15	163.64	945.14	2,880,000.00 *gal*
Application event totals		127.46	9.15	163.64	945.14	

TC8 - 06/19/2023: Corn, silage

Field name: TC8

Crop: Corn, silage Plant date: 06/19/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

TC8 - 06/19/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well #1 Office	Ground water	0.24	0.00	0.00	24.97	1,479,600.00 *gal*
Ag Well #2 East	Ground water	0.25	0.00	0.00	233.07	1,553,580.00 *gal*
Ag Well #3 South	Ground water	0.71	0.00	0.00	28.44	1,516,590.00 *gal*
Application event totals		1.20	0.00	0.00	286.48	

07/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well #1 Office	Ground water	0.24	0.00	0.00	24.97	1,479,600.00 *gal*
Ag Well #2 East	Ground water	0.25	0.00	0.00	233.07	1,553,580.00 *gal*
Ag Well #3 South	Ground water	0.71	0.00	0.00	28.44	1,516,590.00 *gal*
Application event totals		1.20	0.00	0.00	286.48	

07/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	6.18	1.51	12.56	68.62	950,400.00 *gal*
Ag Well #1 Office	Ground water	0.13	0.00	0.00	13.37	792,000.00 *gal*
Ag Well #2 East	Ground water	0.13	0.00	0.00	124.76	831,600.00 *gal*
Ag Well #3 South	Ground water	0.38	0.00	0.00	15.22	811,800.00 *gal*
Application event totals		6.83	1.51	12.56	221.97	

08/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	6.18	1.51	12.56	68.62	950,400.00 *gal*
Ag Well #1 Office	Ground water	0.13	0.00	0.00	13.37	792,000.00 *gal*
Ag Well #2 East	Ground water	0.13	0.00	0.00	124.76	831,600.00 *gal*
Ag Well #3 South	Ground water	0.38	0.00	0.00	15.22	811,800.00 *gal*
Application event totals		6.83	1.51	12.56	221.97	

TC8 - 06/19/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	53.82	4.23	63.81	294.07	950,400.00 *gal*
Ag Well #1 Office	Ground water	0.13	0.00	0.00	13.37	792,000.00 *gal*
Ag Well #2 East	Ground water	0.13	0.00	0.00	124.76	831,600.00 *gal*
Ag Well #3 South	Ground water	0.38	0.00	0.00	15.22	811,800.00 *gal*
Application event totals		54.47	4.23	63.81	447.42	

09/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well #1 Office	Ground water	0.20	0.00	0.00	20.86	1,236,000.00 *gal*
Ag Well #2 East	Ground water	0.21	0.00	0.00	194.70	1,297,800.00 *gal*
Ag Well #3 South	Ground water	0.59	0.00	0.00	23.76	1,266,900.00 *gal*
Application event totals		1.00	0.00	0.00	239.32	

TC9 - 06/20/2023: Corn, silage

Field name: TC9

Crop: Corn, silage

Plant date: 06/20/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well #1 Office	Ground water	0.34	0.00	0.00	36.05	360,000.00 *gal*
Ag Well #2 East	Ground water	0.36	0.00	0.00	336.47	378,000.00 *gal*
Ag Well #3 South	Ground water	1.02	0.00	0.00	41.06	369,000.00 *gal*
Application event totals		1.73	0.00	0.00	413.58	

TC9 - 06/20/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well #1 Office	Ground water	0.34	0.00	0.00	36.05	360,000.00 *gal*
Ag Well #2 East	Ground water	0.36	0.00	0.00	336.47	378,000.00 *gal*
Ag Well #3 South	Ground water	1.02	0.00	0.00	41.06	369,000.00 *gal*
Application event totals		1.73	0.00	0.00	413.58	

07/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	8.90	2.17	18.07	98.70	230,400.00 *gal*
Ag Well #1 Office	Ground water	0.18	0.00	0.00	19.23	192,000.00 *gal*
Ag Well #2 East	Ground water	0.19	0.00	0.00	179.45	201,600.00 *gal*
Ag Well #3 South	Ground water	0.55	0.00	0.00	21.90	196,800.00 *gal*
Application event totals		9.82	2.17	18.07	319.27	

08/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	8.90	2.17	18.07	98.70	230,400.00 *gal*
Ag Well #1 Office	Ground water	0.18	0.00	0.00	19.23	192,000.00 *gal*
Ag Well #2 East	Ground water	0.19	0.00	0.00	179.45	201,600.00 *gal*
Ag Well #3 South	Ground water	0.55	0.00	0.00	21.90	196,800.00 *gal*
Application event totals		9.82	2.17	18.07	319.27	

08/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	77.42	6.09	91.78	422.99	230,400.00 *gal*
Ag Well #1 Office	Ground water	0.18	0.00	0.00	19.23	192,000.00 *gal*
Ag Well #2 East	Ground water	0.19	0.00	0.00	179.45	201,600.00 *gal*
Ag Well #3 South	Ground water	0.55	0.00	0.00	21.90	196,800.00 *gal*
Application event totals		78.34	6.09	91.78	643.57	

TC9 - 06/20/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well #1 Office	Ground water	0.27	0.00	0.00	28.00	279,600.00 *gal*
Ag Well #2 East	Ground water	0.28	0.00	0.00	261.33	293,580.00 *gal*
Ag Well #3 South	Ground water	0.80	0.00	0.00	31.89	286,590.00 *gal*
Application event totals		1.35	0.00	0.00	321.21	

TC9A - 10/22/2022: Oats, silage-soft dough

Field name: TC9A

Crop: Oats, silage-soft dough Plant date: 10/22/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/08/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Manure	Corral solids	200.03	68.04	167.13	5,435.08	500.00 *ton*
Application event totals		200.03	68.04	167.13	5,435.08	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
01/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	45.38	3.26	58.26	336.47	576,000.00 *gal*
Application event totals		45.38	3.26	58.26	336.47	

TC9A - 06/21/2023: Corn, silage

Field name: TC9A

Crop: Corn, silage Plant date: 06/21/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

TC9A - 06/21/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well #1 Office	Ground water	0.37	0.00	0.00	38.43	1,279,200.00 *gal*
Ag Well #2 East	Ground water	0.39	0.00	0.00	358.68	1,343,160.00 *gal*
Ag Well #3 South	Ground water	1.09	0.00	0.00	43.77	1,311,180.00 *gal*
Application event totals		1.85	0.00	0.00	440.87	

07/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well #1 Office	Ground water	0.37	0.00	0.00	38.43	1,279,200.00 *gal*
Ag Well #2 East	Ground water	0.39	0.00	0.00	358.68	1,343,160.00 *gal*
Ag Well #3 South	Ground water	1.09	0.00	0.00	43.77	1,311,180.00 *gal*
Application event totals		1.85	0.00	0.00	440.87	

07/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	10.01	2.44	20.33	111.04	864,000.00 *gal*
Ag Well #1 Office	Ground water	0.21	0.00	0.00	21.63	720,000.00 *gal*
Ag Well #2 East	Ground water	0.22	0.00	0.00	201.88	756,000.00 *gal*
Ag Well #3 South	Ground water	0.61	0.00	0.00	24.63	738,000.00 *gal*
Application event totals		11.05	2.44	20.33	359.18	

08/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	10.01	2.44	20.33	111.04	864,000.00 *gal*
Ag Well #1 Office	Ground water	0.21	0.00	0.00	21.63	720,000.00 *gal*
Ag Well #2 East	Ground water	0.22	0.00	0.00	201.88	756,000.00 *gal*
Ag Well #3 South	Ground water	0.61	0.00	0.00	24.63	738,000.00 *gal*
Application event totals		11.05	2.44	20.33	359.18	

TC9A - 06/21/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	87.10	6.85	103.25	475.87	864,000.00 *gal*
Ag Well #1 Office	Ground water	0.21	0.00	0.00	21.63	720,000.00 *gal*
Ag Well #2 East	Ground water	0.22	0.00	0.00	201.88	756,000.00 *gal*
Ag Well #3 South	Ground water	0.61	0.00	0.00	24.63	738,000.00 *gal*
Application event totals		88.14	6.85	103.25	724.01	

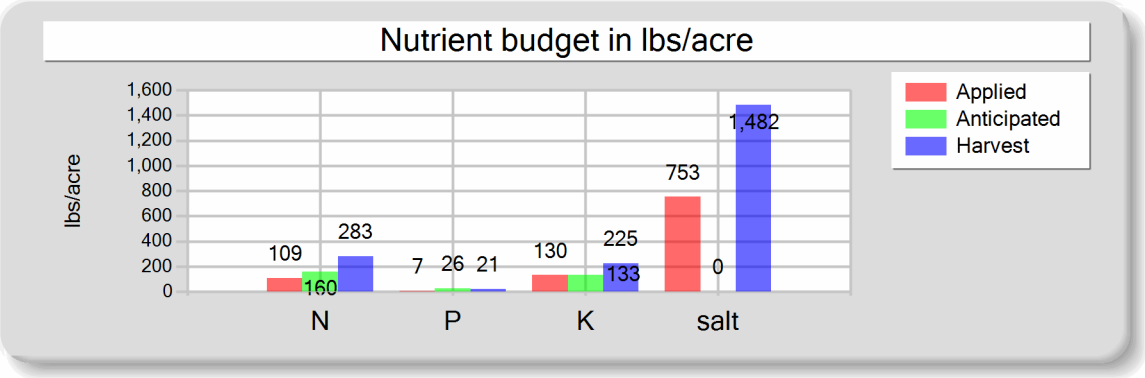
09/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well #1 Office	Ground water	0.32	0.00	0.00	33.64	1,119,600.00 *gal*
Ag Well #2 East	Ground water	0.34	0.00	0.00	313.93	1,175,580.00 *gal*
Ag Well #3 South	Ground water	0.96	0.00	0.00	38.31	1,147,590.00 *gal*
Application event totals		1.62	0.00	0.00	385.87	

B. NUTRIENT BUDGET

TC1 - 10/24/2022: Oats, silage-soft dough

Field name: TC1 Crop: Oats, silage-soft dough Plant date: 10/24/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	101.59	7.30	130.43	753.29
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	108.59	7.30	130.43	753.29
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	282.59	20.53	225.44	1,481.76
Nutrient balance	-174.01	-13.24	-95.01	-728.47
Applied to removed ratio	0.38	0.36	0.58	0.51

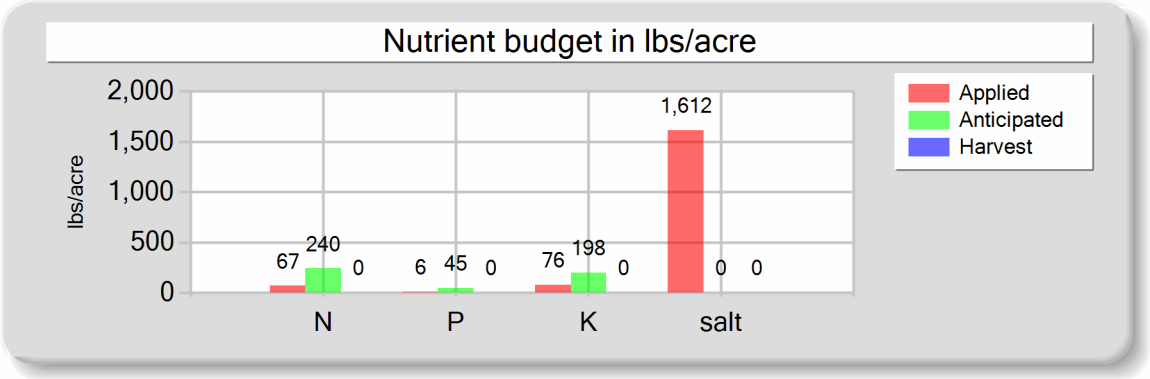
Fresh water applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Process wastewater applied	
1,728,000.00	*gallons*
63.64	*acre-inches*
0.95	*inches/acre*

Total harvests for the crop	
1	*harvests*

TC1 - 06/15/2023: Corn, silage

Field name: TC1 Crop: Corn, silage Plant date: 06/15/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	55.28	6.32	75.64	369.33
Fresh water	5.21	0.00	0.00	1,242.59
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	67.49	6.32	75.64	1,611.91
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	0.00	0.00	0.00	0.00
Nutrient balance	67.49	6.32	75.64	1,611.91
Applied to removed ratio	0.00	0.00	0.00	0.00

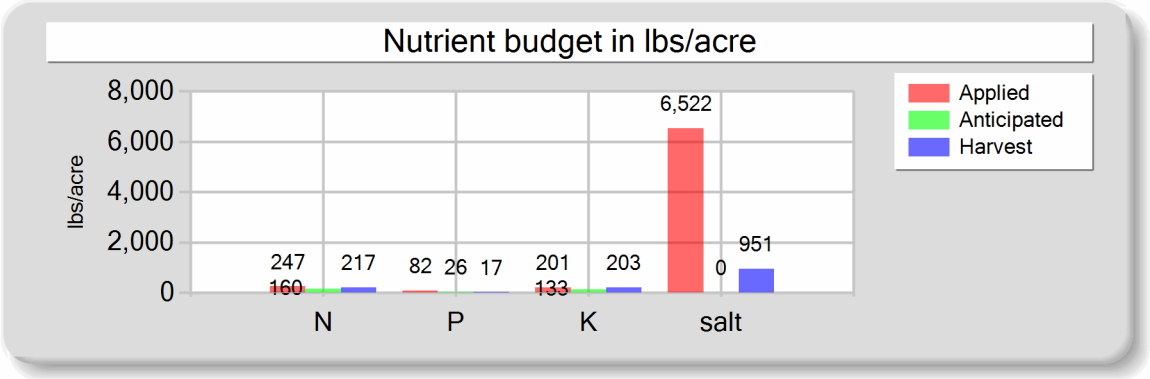
Fresh water applied	
14,855,940.00	*gallons*
547.09	*acre-inches*
8.17	*inches/acre*

Process wastewater applied	
1,958,400.00	*gallons*
72.12	*acre-inches*
1.08	*inches/acre*

Total harvests for the crop	
1	*harvests*

TC10 - 10/22/2022: Oats, silage-soft dough

Field name: TC10 Crop: Oats, silage-soft dough Plant date: 10/22/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	240.04	81.64	200.56	6,522.10
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	247.04	81.64	200.56	6,522.10
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	216.72	16.70	202.86	951.30
Nutrient balance	30.32	64.95	-2.30	5,570.80
Applied to removed ratio	1.14	4.89	0.99	6.86

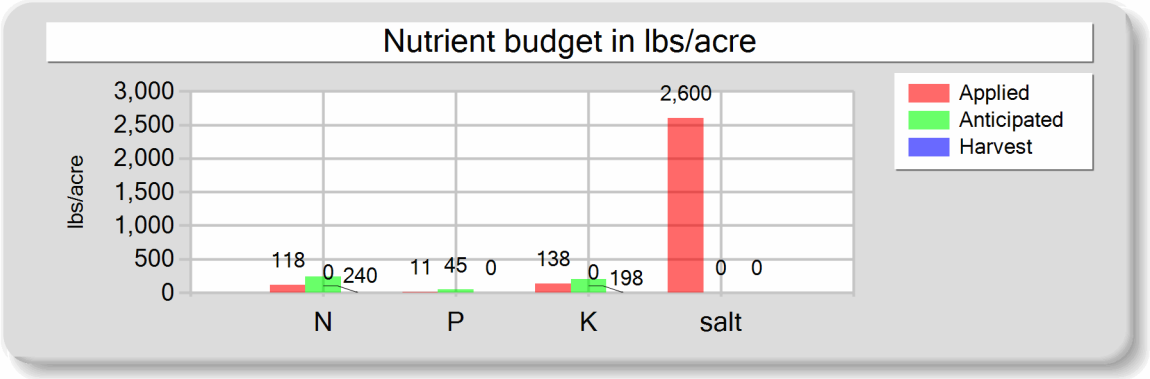
Fresh water applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Process wastewater applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Total harvests for the crop
1 *harvests*

TC10 - 06/15/2023: Corn, silage

Field name: TC10 Crop: Corn, silage Plant date: 06/15/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	102.88	11.26	138.22	670.33
Fresh water	8.08	0.00	0.00	1,929.80
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	117.96	11.26	138.22	2,600.13
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	0.00	0.00	0.00	0.00
Nutrient balance	117.96	11.26	138.22	2,600.13
Applied to removed ratio	0.00	0.00	0.00	0.00

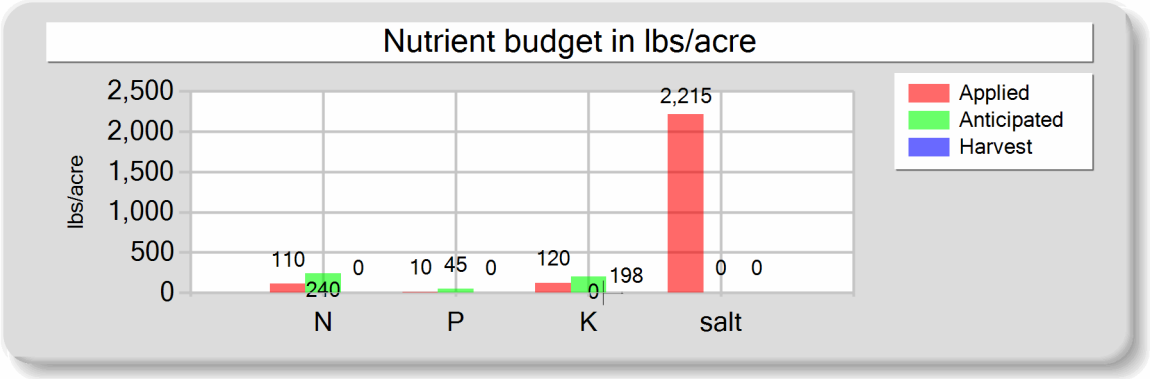
Fresh water applied	
20,317,140.00	*gallons*
748.21	*acre-inches*
12.68	*inches/acre*

Process wastewater applied	
2,937,600.00	*gallons*
108.18	*acre-inches*
1.83	*inches/acre*

Total harvests for the crop	
1	*harvests*

TC2 - 06/16/2023: Corn, silage

Field name: TC2 Crop: Corn, silage Plant date: 06/16/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	89.26	9.77	119.93	581.61
Fresh water	6.42	0.00	0.00	1,633.78
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	109.68	9.77	119.93	2,215.40
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	0.00	0.00	0.00	0.00
Nutrient balance	109.68	9.77	119.93	2,215.40
Applied to removed ratio	0.00	0.00	0.00	0.00

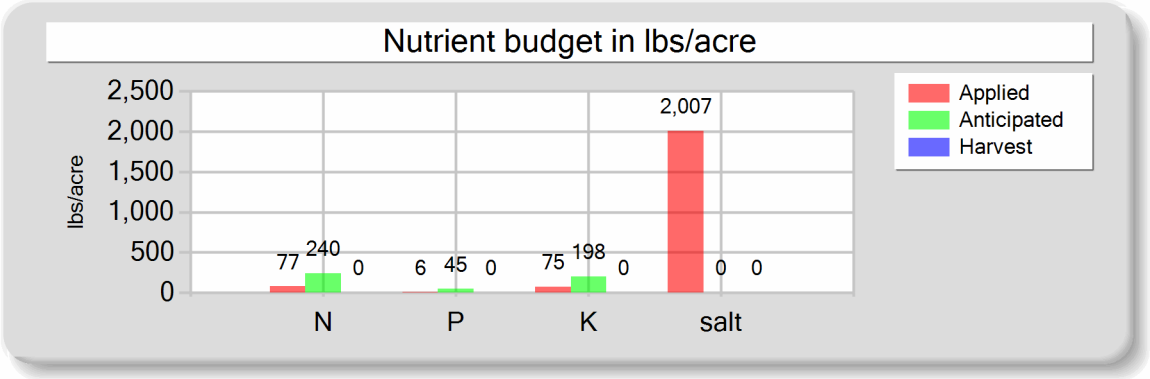
Fresh water applied	
7,921,200.00	*gallons*
291.71	*acre-inches*
10.42	*inches/acre*

Process wastewater applied	
1,209,600.00	*gallons*
44.55	*acre-inches*
1.59	*inches/acre*

Total harvests for the crop	
1	*harvests*

TC3 - 06/16/2023: Corn, silage

Field name: TC3 Crop: Corn, silage Plant date: 06/16/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	55.79	6.11	74.95	363.51
Fresh water	6.88	0.00	0.00	1,643.18
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	76.67	6.11	74.95	2,006.69
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	0.00	0.00	0.00	0.00
Nutrient balance	76.67	6.11	74.95	2,006.69
Applied to removed ratio	0.00	0.00	0.00	0.00

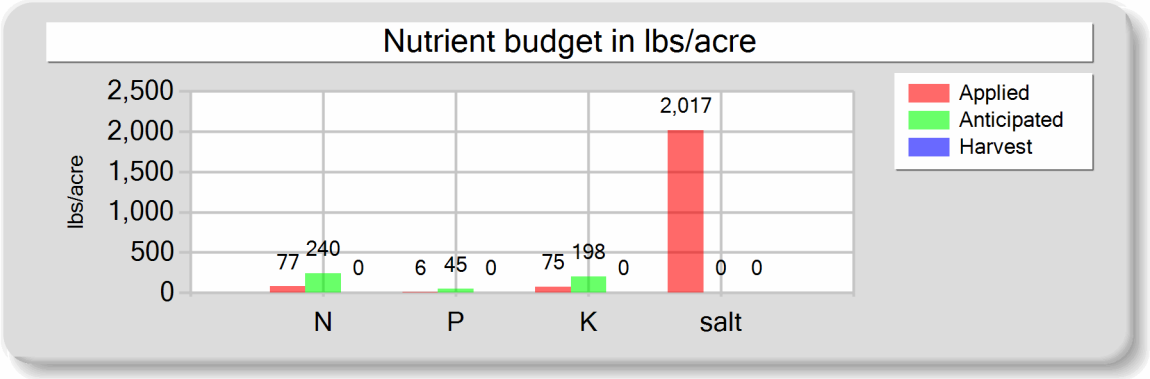
Fresh water applied	
7,623,540.00	*gallons*
280.75	*acre-inches*
10.80	*inches/acre*

Process wastewater applied	
702,000.00	*gallons*
25.85	*acre-inches*
0.99	*inches/acre*

Total harvests for the crop	
1	*harvests*

TC4 - 06/17/2023: Corn, silage

Field name: TC4 Crop: Corn, silage Plant date: 06/17/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	55.79	6.11	74.95	363.51
Fresh water	6.93	0.00	0.00	1,653.55
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	76.72	6.11	74.95	2,017.06
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	0.00	0.00	0.00	0.00
Nutrient balance	76.72	6.11	74.95	2,017.06
Applied to removed ratio	0.00	0.00	0.00	0.00

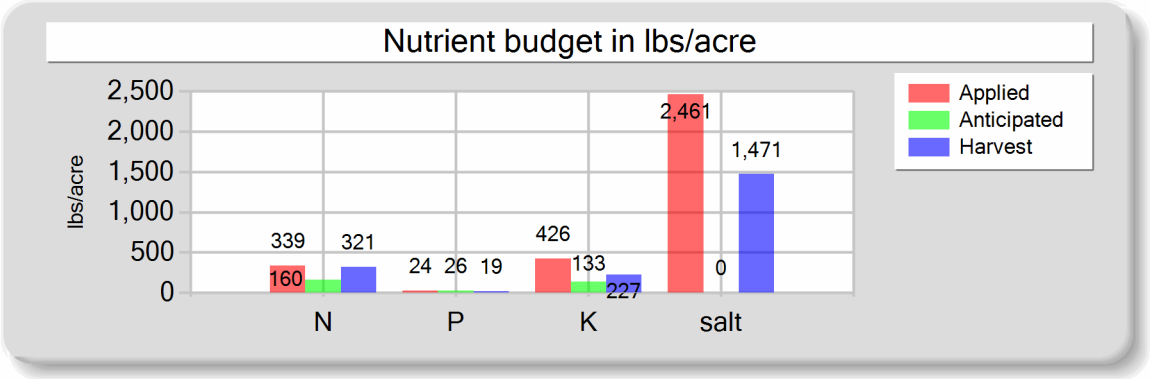
Fresh water applied	
7,966,710.00	*gallons*
293.39	*acre-inches*
10.87	*inches/acre*

Process wastewater applied	
729,000.00	*gallons*
26.85	*acre-inches*
0.99	*inches/acre*

Total harvests for the crop	
1	*harvests*

TC5 - 10/26/2022: Oats, silage-soft dough

Field name: TC5 Crop: Oats, silage-soft dough Plant date: 10/26/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	331.82	23.83	426.02	2,460.52
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	338.82	23.83	426.02	2,460.52
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	321.09	18.56	226.51	1,471.18
Nutrient balance	17.73	5.27	199.51	989.34
Applied to removed ratio	1.06	1.28	1.88	1.67

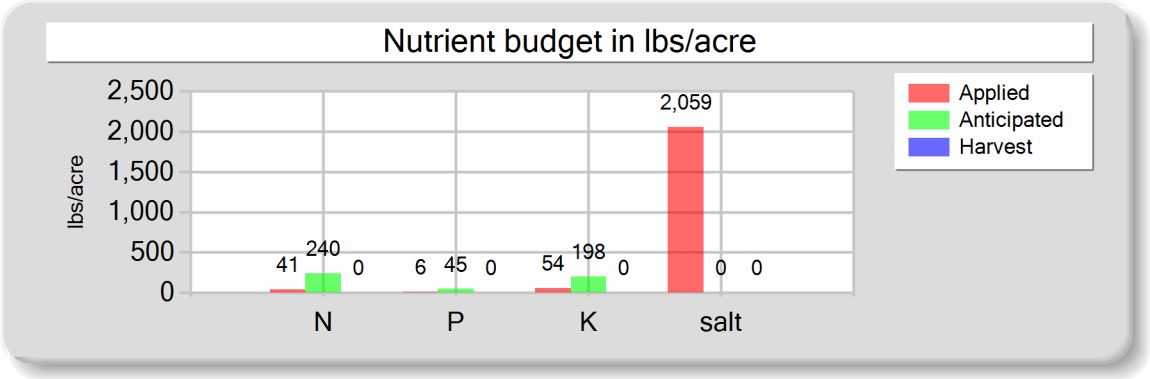
Fresh water applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Process wastewater applied
2,527,280.00 *gallons*
93.07 *acre-inches*
3.10 *inches/acre*

Total harvests for the crop
1 *harvests*

TC5 - 06/17/2023: Corn, silage

Field name: TC5 Crop: Corn, silage Plant date: 06/17/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	26.69	6.50	54.22	296.09
Fresh water	7.39	0.00	0.00	1,763.22
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	41.07	6.50	54.22	2,059.32
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	0.00	0.00	0.00	0.00
Nutrient balance	41.07	6.50	54.22	2,059.32
Applied to removed ratio	0.00	0.00	0.00	0.00

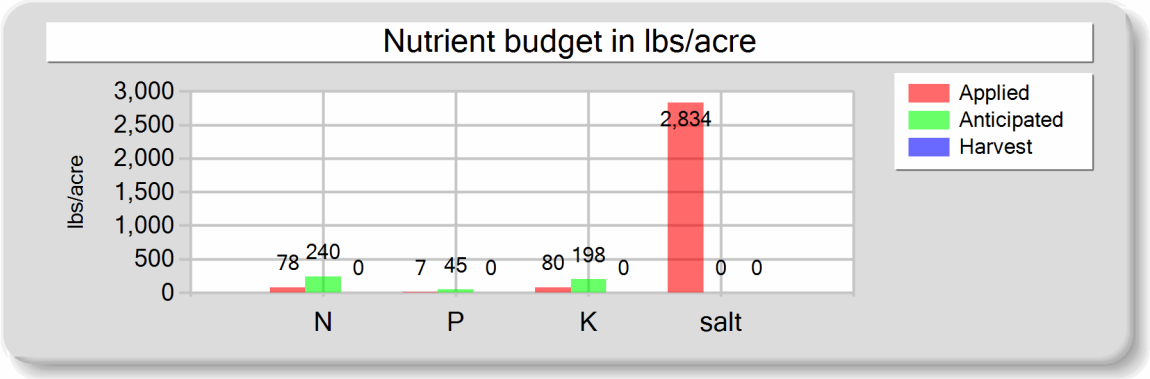
Fresh water applied	
9,439,020.00	*gallons*
347.61	*acre-inches*
11.59	*inches/acre*

Process wastewater applied	
1,382,400.00	*gallons*
50.91	*acre-inches*
1.70	*inches/acre*

Total harvests for the crop	
1	*harvests*

TC6 - 06/18/2023: Corn, silage

Field name: TC6 Crop: Corn, silage Plant date: 06/18/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	59.51	6.51	79.95	387.74
Fresh water	4.65	0.00	0.00	2,446.22
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	78.16	6.51	79.95	2,833.96
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	0.00	0.00	0.00	0.00
Nutrient balance	78.16	6.51	79.95	2,833.96
Applied to removed ratio	0.00	0.00	0.00	0.00

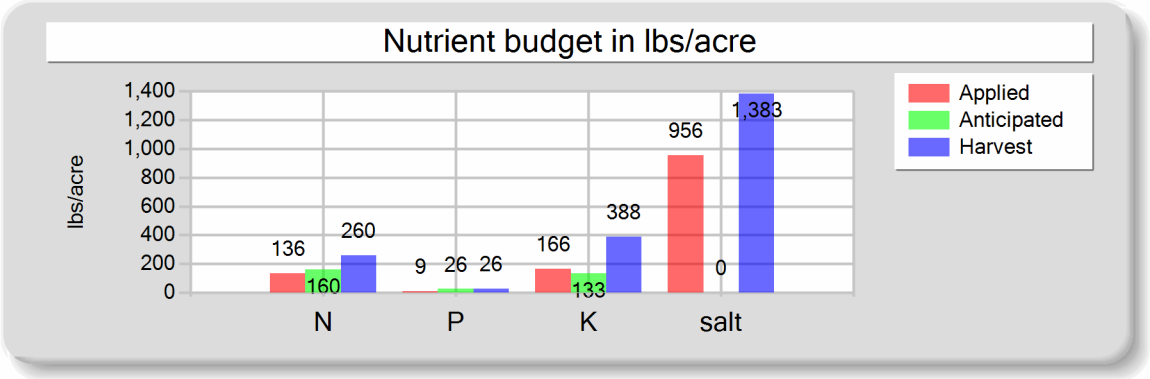
Fresh water applied	
4,846,200.00	*gallons*
178.47	*acre-inches*
11.90	*inches/acre*

Process wastewater applied	
432,000.00	*gallons*
15.91	*acre-inches*
1.06	*inches/acre*

Total harvests for the crop	
1	*harvests*

TC7 - 10/24/2022: Oats, silage-soft dough

Field name: TC7 Crop: Oats, silage-soft dough Plant date: 10/24/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	128.91	9.26	165.50	955.88
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	135.91	9.26	165.50	955.88
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	259.69	26.31	387.83	1,383.48
Nutrient balance	-123.78	-17.05	-222.32	-427.60
Applied to removed ratio	0.52	0.35	0.43	0.69

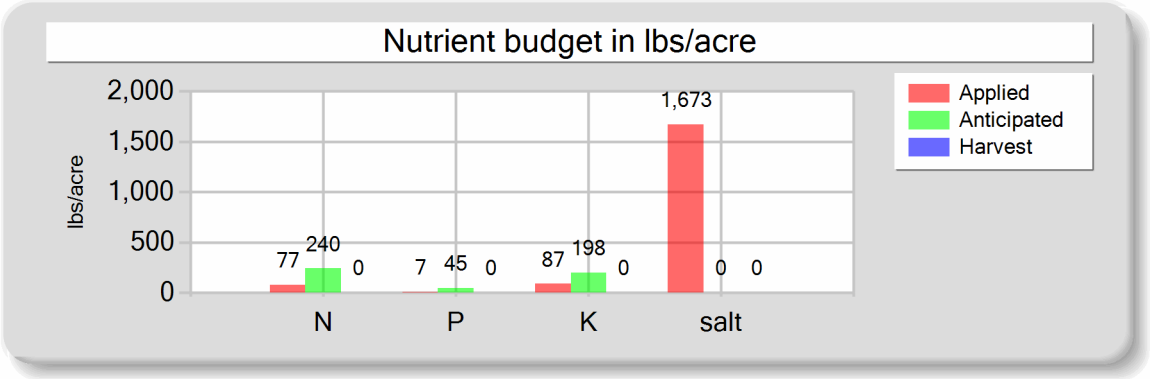
Fresh water applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Process wastewater applied	
2,880,000.00	*gallons*
106.06	*acre-inches*
1.21	*inches/acre*

Total harvests for the crop	
1	*harvests*

TC7 - 06/18/2023: Corn, silage

Field name: TC7 Crop: Corn, silage Plant date: 06/18/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	64.92	7.11	87.22	422.99
Fresh water	5.24	0.00	0.00	1,250.13
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	77.15	7.11	87.22	1,673.13
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	0.00	0.00	0.00	0.00
Nutrient balance	77.15	7.11	87.22	1,673.13
Applied to removed ratio	0.00	0.00	0.00	0.00

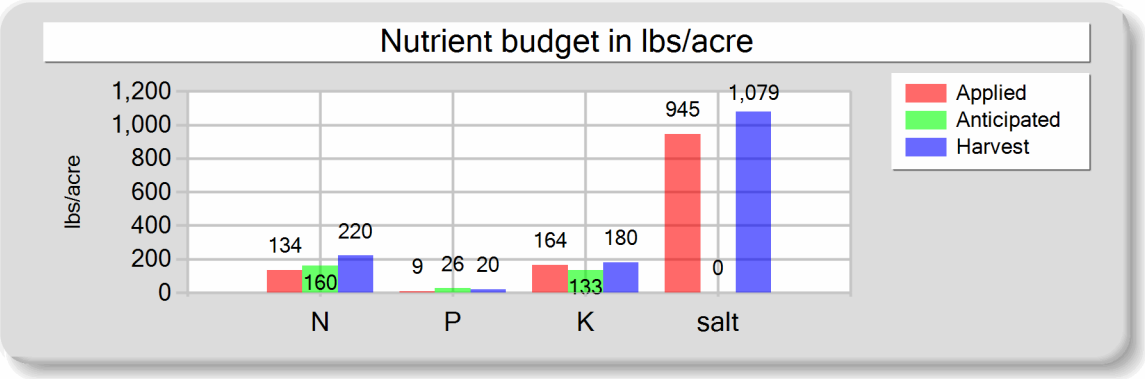
Fresh water applied	
19,630,800.00	*gallons*
722.94	*acre-inches*
8.22	*inches/acre*

Process wastewater applied	
2,764,800.00	*gallons*
101.82	*acre-inches*
1.16	*inches/acre*

Total harvests for the crop	
1	*harvests*

TC8 - 10/23/2022: Oats, silage-soft dough

Field name: TC8 Crop: Oats, silage-soft dough Plant date: 10/23/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	127.46	9.15	163.64	945.14
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	134.46	9.15	163.64	945.14
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	220.32	19.73	180.50	1,079.46
Nutrient balance	-85.86	-10.58	-16.85	-134.31
Applied to removed ratio	0.61	0.46	0.91	0.88

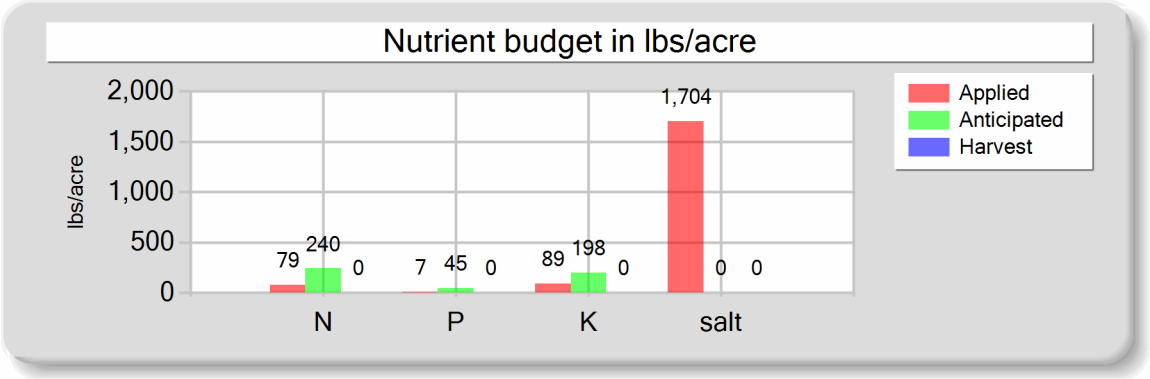
Fresh water applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Process wastewater applied	
2,880,000.00	*gallons*
106.06	*acre-inches*
1.19	*inches/acre*

Total harvests for the crop	
1	*harvests*

TC8 - 06/19/2023: Corn, silage

Field name: TC8 Crop: Corn, silage Plant date: 06/19/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	66.19	7.24	88.94	431.31
Fresh water	5.33	0.00	0.00	1,272.33
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	78.52	7.24	88.94	1,703.64
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	0.00	0.00	0.00	0.00
Nutrient balance	78.52	7.24	88.94	1,703.64
Applied to removed ratio	0.00	0.00	0.00	0.00

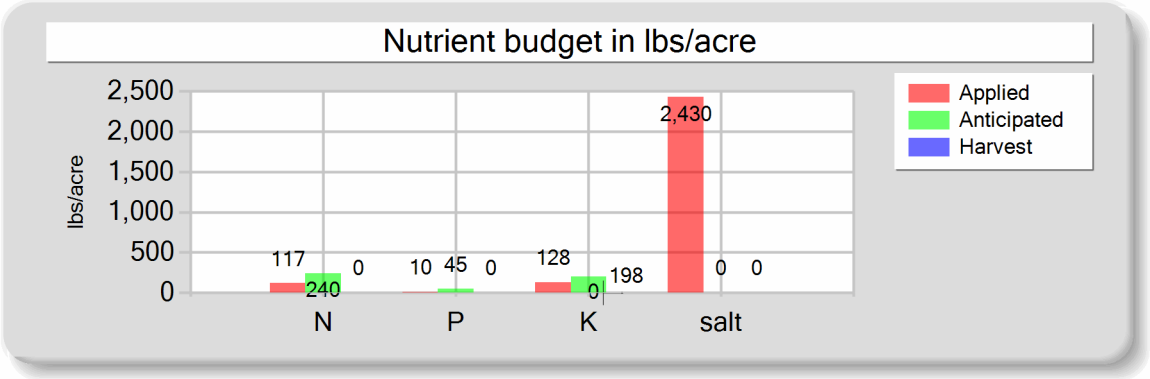
Fresh water applied	
20,206,440.00	*gallons*
744.13	*acre-inches*
8.36	*inches/acre*

Process wastewater applied	
2,851,200.00	*gallons*
105.00	*acre-inches*
1.18	*inches/acre*

Total harvests for the crop	
1	*harvests*

TC9 - 06/20/2023: Corn, silage

Field name: TC9 Crop: Corn, silage Plant date: 06/20/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	95.21	10.42	127.92	620.39
Fresh water	7.58	0.00	0.00	1,810.09
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	116.79	10.42	127.92	2,430.48
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	0.00	0.00	0.00	0.00
Nutrient balance	116.79	10.42	127.92	2,430.48
Applied to removed ratio	0.00	0.00	0.00	0.00

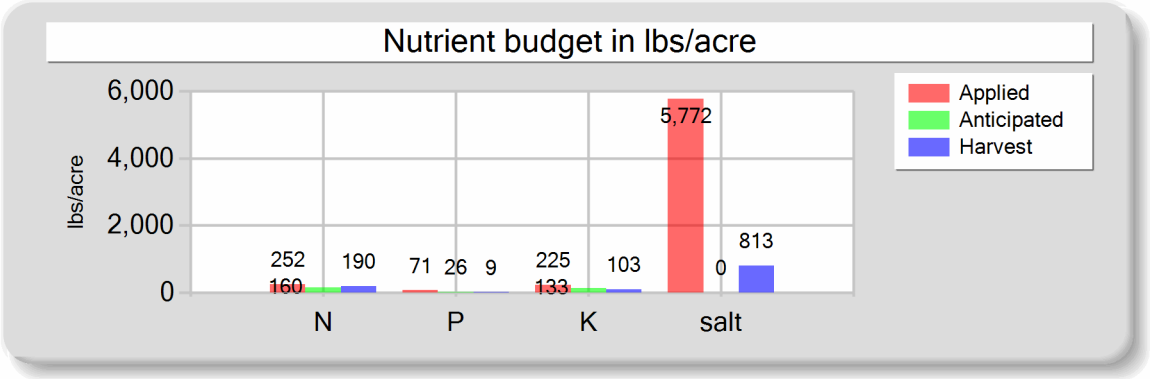
Fresh water applied	
4,844,970.00	*gallons*
178.42	*acre-inches*
11.89	*inches/acre*

Process wastewater applied	
691,200.00	*gallons*
25.45	*acre-inches*
1.70	*inches/acre*

Total harvests for the crop	
1	*harvests*

TC9A - 10/22/2022: Oats, silage-soft dough

Field name: TC9A | Crop: Oats, silage-soft dough | Plant date: 10/22/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	200.03	68.04	167.13	5,435.08
Process wastewater	45.38	3.26	58.26	336.47
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	252.41	71.30	225.39	5,771.55
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	189.59	8.96	102.56	813.26
Nutrient balance	62.82	62.33	122.83	4,958.29
Applied to removed ratio	1.33	7.96	2.20	7.10

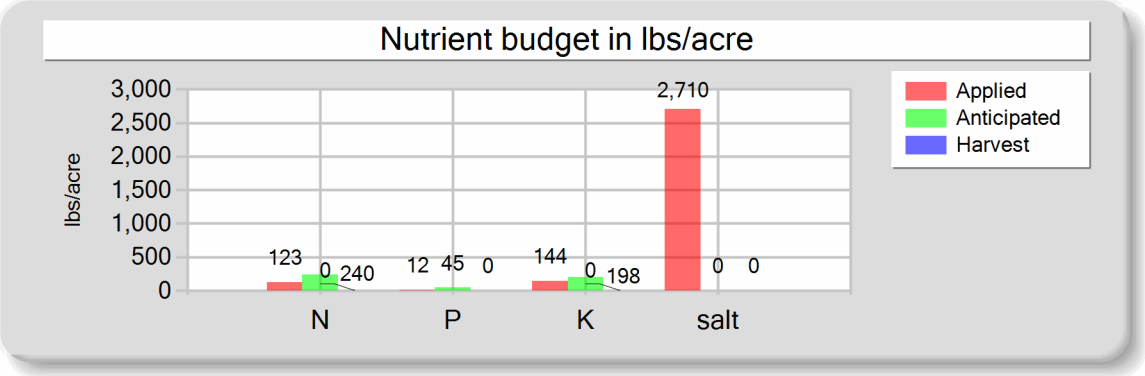
Fresh water applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Process wastewater applied	
576,000.00	*gallons*
21.21	*acre-inches*
0.42	*inches/acre*

Total harvests for the crop	
1	*harvests*

TC9A - 06/21/2023: Corn, silage

Field name: TC9A | Crop: Corn, silage | Plant date: 06/21/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	107.11	11.72	143.91	697.94
Fresh water	8.43	0.00	0.00	2,012.06
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	122.54	11.72	143.91	2,709.99
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	0.00	0.00	0.00	0.00
Nutrient balance	122.54	11.72	143.91	2,709.99
Applied to removed ratio	0.00	0.00	0.00	0.00

Fresh water applied	
17,951,850.00	*gallons*
661.11	*acre-inches*
13.22	*inches/acre*

Process wastewater applied	
2,592,000.00	*gallons*
95.45	*acre-inches*
1.91	*inches/acre*

Total harvests for the crop	
1	*harvests*

NUTRIENT ANALYSES

A. MANURE ANALYSES

Manure 2101306-01

Sample and source description: Manure 2101306-01

Sample date: 10/13/2022 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 34.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	15,200.00	5,170.00	12,700.00	110,000.00	5,200.00	2,400.00	141.00	1,700.00		41.30
DL	0.50	0.20	0.20	5.00	5.00	5.00	0.10	0.10		0.10

B. PROCESS WASTEWATER ANALYSES

Lagoon Water 2302040-01

Sample and source description: Lagoon Water 2302040-01

Sample date: 01/31/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	472.00	472.00			33.90	606.00								4,830.00	3,500
DL	0.60	0.60			0.10	10.00								1.00	1

Lagoon Water 2303402-01

Sample and source description: Lagoon Water 2303402-01

Sample date: 03/27/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	380.00	380.00			31.90	607.00								4,570.00	2,700
DL	0.60	5.00			0.10	10.00								1.00	1

Lagoon Water 2307249-01

Sample and source description: Lagoon Water 2307249-01

Sample date: 07/16/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	69.40	69.40			16.90	141.00								1,600.00	770
DL	0.60	0.60			0.10	10.00								1.00	1

Lagoon Water 2310006-01

Sample and source description: Lagoon Water 2310006-01

Sample date: 09/29/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	604.00	422.00			47.50	716.00								5,760.00	3,300
DL	1.20	0.60			0.10	10.00								1.00	1

C. FRESH WATER ANALYSES

Ag Well #1 Office

2311037-01

Sample description: 2311037-01

Sample date: 11/03/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	1.72										231.00	180
DL	1.40										1.00	1

Ag Well #2 East

Ag Well #2 East

2311037-02

Sample description: 2311037-02

Sample date: 11/03/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	1.73										240.00	1,600
DL	1.40										1.00	1

Ag Well #3 South

2311037-03

Sample description: 2311037-03

Sample date: 11/03/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	4.99										349.00	200
DL	1.40										1.00	1

Domestic Well

2310491-01

Sample description: 2310491-01

Sample date: 10/25/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	3.71										300.00	
DL	0.40										1.00	

D. SOIL ANALYSES

No soil analyses entered.

E. PLANT TISSUE ANALYSES

TC1 - 10/24/2022: Oats, silage-soft dough

2305296-01

Sample and source description: 2305296-01

Sample date: 05/16/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 62.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	26,700.00	1,940.00	21,300.00		14.00
DL	1.00	0.50	0.50		0.10

TC10 - 10/22/2022: Oats, silage-soft dough

2305297-03

Sample and source description: 2305297-03

Sample date: 05/16/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 77.5 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	34,400.00	2,650.00	32,200.00		15.10
DL	1.00	0.50	0.50		0.10

TC5 - 10/26/2022: Oats, silage-soft dough

TC5 - 10/26/2022: Oats, silage-soft dough

2305296-05

Sample and source description: 2305296-05

Sample date: 05/16/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 58.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	27,500.00	1,590.00	19,400.00		12.60
DL	2.50	0.50	0.50		0.10

TC7 - 10/24/2022: Oats, silage-soft dough

2305296-07

Sample and source description: 2305296-07

Sample date: 05/16/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 59.5 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	22,900.00	2,320.00	34,200.00		12.20
DL	1.00	0.50	0.50		0.10

TC8 - 10/23/2022: Oats, silage-soft dough

2305296-08

Sample and source description: 2305296-08

Sample date: 05/16/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 68.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	24,900.00	2,230.00	20,400.00		12.20
DL	2.50	0.50	0.50		0.10

TC9A - 10/22/2022: Oats, silage-soft dough

2305297-02

Sample and source description: 2305297-02

Sample date: 05/26/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 81.5 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	36,600.00	1,730.00	19,800.00		15.70
DL	0.50	0.50	0.50		0.10

F. SUBSURFACE (TILE) DRAINAGE ANALYSES

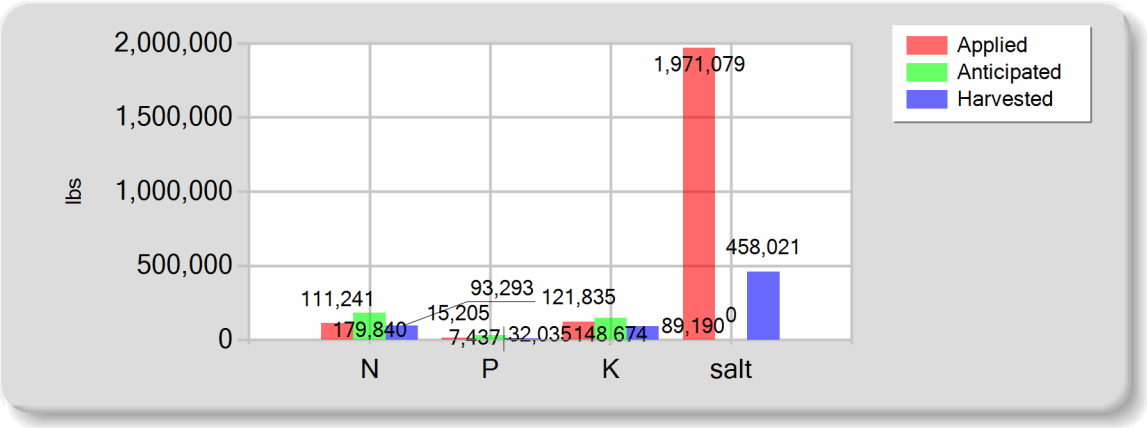
No subsurface (tile) drainage analyses entered.

NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

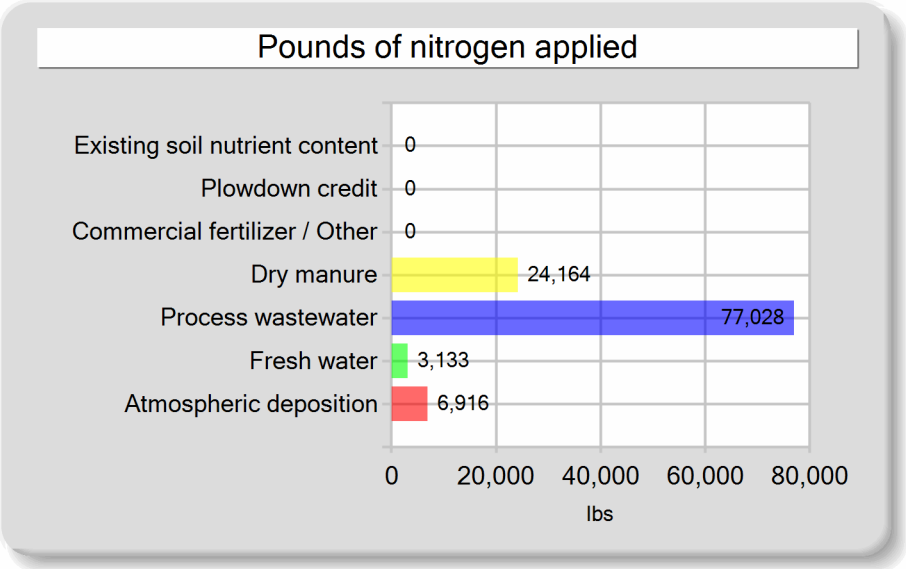
A. SUMMARY OF NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

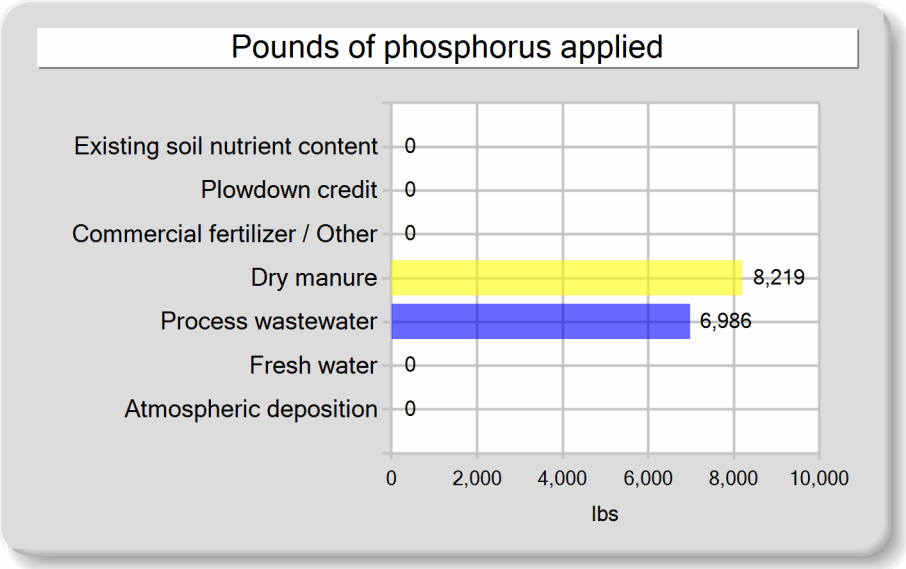
	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	24,163.87	8,218.89	20,189.55	656,557.66
Process wastewater	77,027.96	6,986.22	101,645.68	543,701.85
Fresh water	3,133.43	0.00	0.00	770,819.49
Atmospheric deposition	6,916.00	0.00	0.00	0.00
Total nutrients applied	111,241.26	15,205.11	121,835.22	1,971,079.00
Anticipated crop nutrient removal	179,840.00	32,034.80	148,674.40	0.00
Actual crop nutrient removal	93,292.72	7,436.97	89,190.09	458,020.72
Nutrient balance	17,948.54	7,768.15	32,645.13	1,513,058.28
Applied to removed ratio	1.19	2.04	1.37	4.30

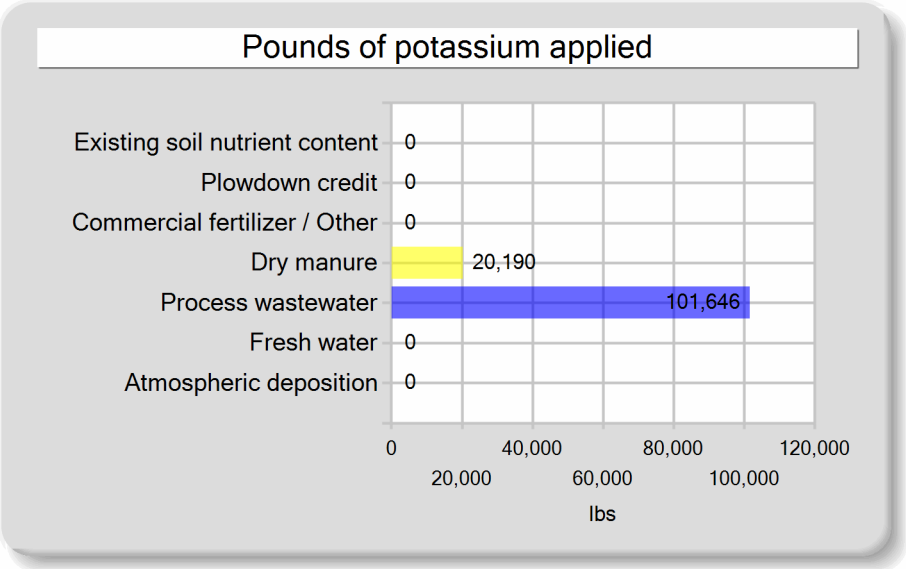
B. POUNDS OF NUTRIENT APPLIED VS. CROP REMOVAL

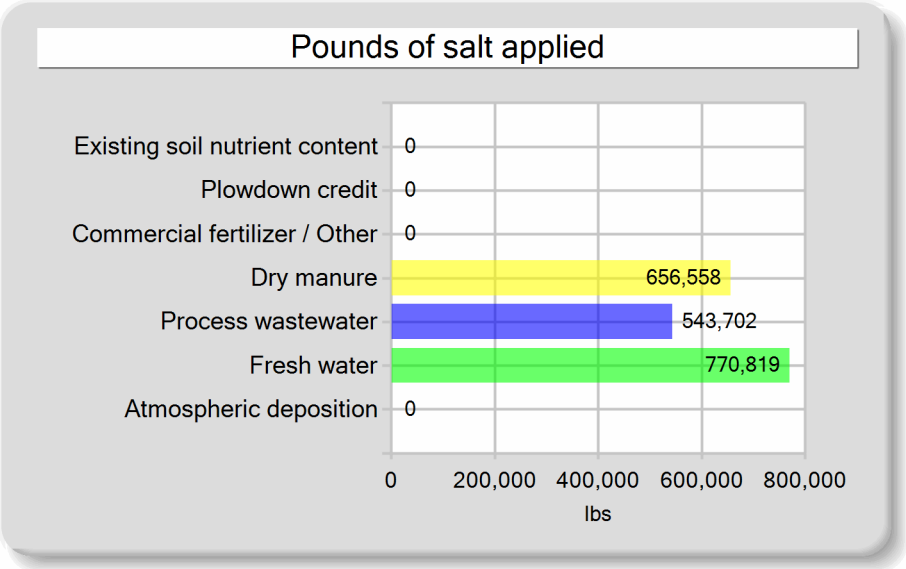


C. POUNDS OF NUTRIENT APPLIED BY MATERIAL TYPE









EXCEPTION REPORTING

A. MANURE, PROCESS WASTEWATER, AND OTHER DAIRY WASTE DISCHARGES

The following is a summary of all manure and process wastewater discharges from the production area to surface water or to land areas (land application areas or otherwise) when not in accordance with the facility's Nutrient Management Plan.

No manure or process wastewater discharges occurred during the reporting period.

B. STORM WATER DISCHARGES

The following is a summary of all storm water discharges from the production area to surface water during the reporting period when not in accordance with the facility 's Nutrient Management Plan.

No stormwater discharges occurred during the reporting period.

C. LAND APPLICATION AREA TO SURFACE WATER DISCHARGES

The following is a summary of all discharges from the land application area to surface water that have occurred during the reporting period when not in accordance with the facility's Nutrient Management Plan.

No land application area to surface water discharges occurred during the reporting period.

NUTRIENT MANAGEMENT PLAN AND EXPORT AGREEMENT STATEMENTS

A. NUTRIENT MANAGEMENT PLAN STATEMENTS

Was the facility's NMP updated in the reporting period?	No
Was the facility's NMP developed by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes
Was the facility's NMP approved by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes

B. EXPORT AGREEMENT STATEMENT

Are there any written agreements with third parties to receive manure or process wastewater that are new or were revised within the reporting period?	No

ADDITIONAL NOTES

A. NOTES

*Field TC1-TC10 Corn Silage was not sample in 2023, we will be sure to sample in 2024.

Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

CERTIFICATION

A. OWNER AND/OR OPERATOR CERTIFICATION

I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and all attachments and that, based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment.

X	
SIGNATURE OF OWNER OF FACILITY	SIGNATURE OF OPERATOR OF FACILITY
Robert Eric Veneman	SAME AS OWNER
PRINT OR TYPE NAME	PRINT OR TYPE NAME
7/15/2024	7/15/2024
DATE	DATE

ATTACHMENTS

A. REQUIRED ATTACHMENTS

The following lists the required documents that should be attached to the Annual Report when submitted .

Annual Dairy Facility Assessment	N/A
Provide an Annual Dairy Facility Assessment (an update to the Preliminary Dairy Facility Assessment in Attachment A) for each reporting period. On the PDFA Final page, click on the ADFA Report button to generate an ADFA report after updating information as needed .	

Manure/Process Wastewater Tracking Manifests	See Attachment
Provide copies of all manure/process wastewater tracking manifests for the reporting period, signed by both the owner/operator and the hauler.	

Corrective Actions Documents	N/A
Provide records documenting any corrective actions taken to correct deficiencies noted as a result of the inspections required in the Monitoring Requirements of the General Order. Deficiencies not corrected in 30 days must be accompanied by an explanation of the factors preventing immediate correction.	

Groundwater Monitoring	N/A
Dischargers that monitor supply wells or subsurface (tile) drainage systems, or that have monitoring well systems must submit monitoring results as directed in the General Order, Groundwater Reporting Section starting on page MRP-13.	

Storm Water Monitoring	N/A
Dischargers that are required to monitor storm water more frequently than required in the General Order must submit monitoring results as directed in the General Order, Storm Water Reporting Section on page MRP-14.	